



36th IEEE/IFIP Network Operations and Management Symposium (NOMS 2023)



LALISA: Adaptive Bitrate Ladder Optimization in HTTP-based Adaptive ...

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Wednesday May 10, 2023 / 03:30 PM - 05:30 PM

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LALISA: Adaptive Bitrate Ladder Optimization in HTTP-based Adaptive Live Streaming

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The 36th IEEE/IFIP Network Operations and Management Symposium (NOMS),
8-12 May- Miami, FL, USA, 2023.



Agenda

- An Introduction on Dynamic Video Streaming
- LALISA: Adaptive Bitrate Ladder Optimization in HTTP-based Adaptive Live Streaming
- LALISA Results
- Future work
- Conclusion

Introduction

 According to a report by Cisco, video streaming accounts for **more than 80%** of all internet traffic.

 As of 2021, YouTube has **over 2 billion monthly active** users who watch **over a billion hours of video every day**.

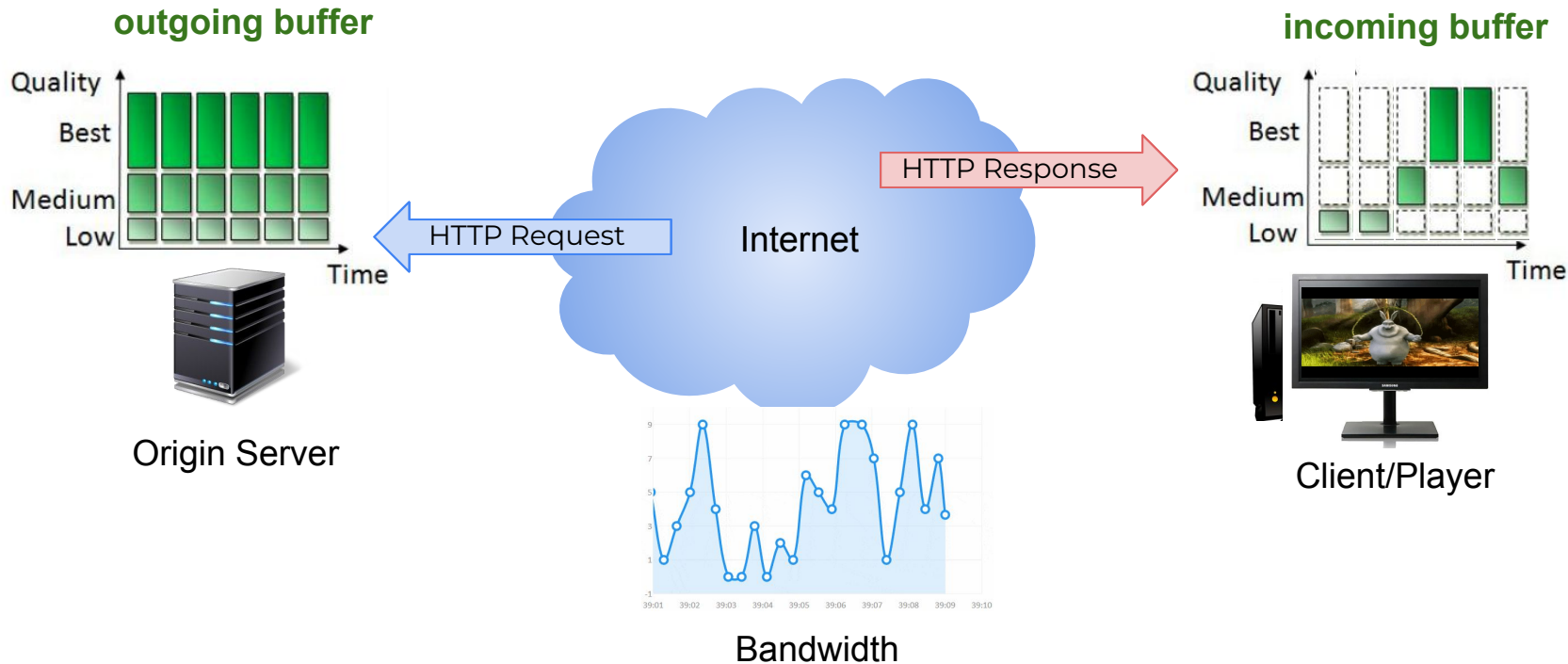
 Netflix has over **209 million subscribers globally**.

 Amazon Prime Video had over **175 million** subscribers worldwide in 2021, with over **200 million hours** of video content streamed **every day**.

 In 2020, Twitch had an average of **30 million daily active users**, who collectively watched over **1 trillion minutes** of content.

In **2021**, the global video streaming market was valued at **\$50.11 billion** and is expected to reach **\$184.27 billion** by **2027**, growing at a **CAGR of 20.4%** from **2020 to 2027**.

Dynamic Adaptive Video Streaming In A Glance



Main Components In Video Streaming Path

Live Video Source



Video on Demand
(VOD)

ABR Encoder
Encoding & Packaging

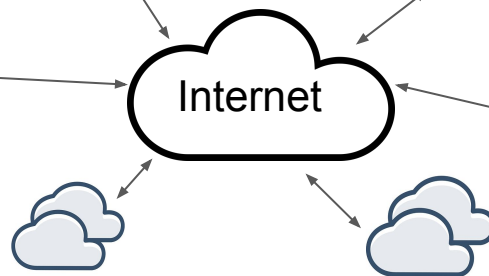
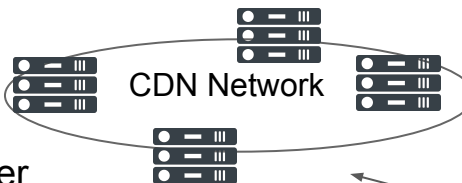


Publishing

Origin
Ingesting



Cloud/Datacenter



Network Core

Network Edge

Base station



ISP



HAS Players



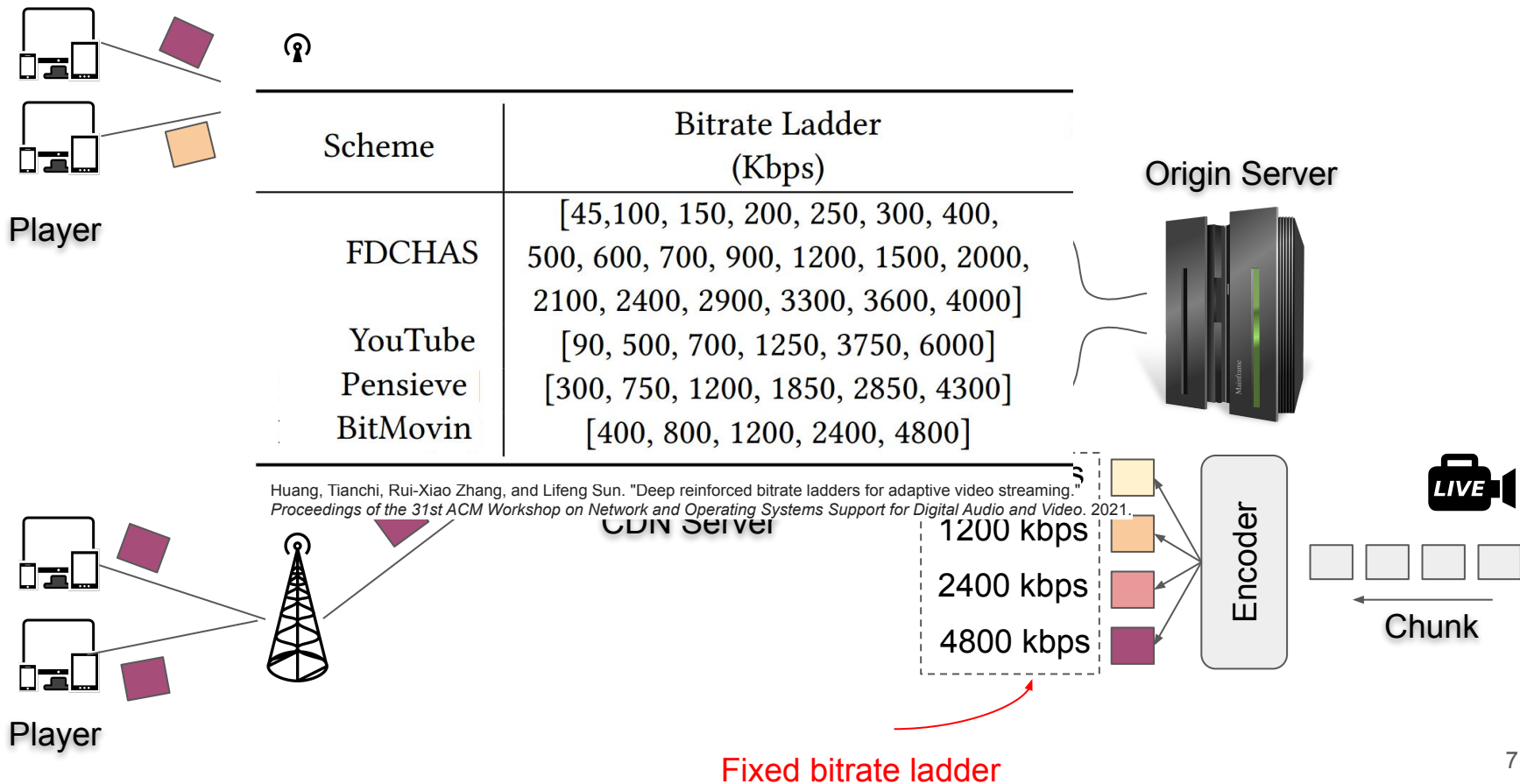
HAS Players

Video Contribution

Video Distribution

Video Consumption

Segment Delivery In Live Streaming



Problem Definition

How to **optimize the bitrate ladder** in the live streaming applications?

How to consider the **content**, **network condition** (i.e., available bandwidth) and **users' demands** (i.e., desired bitrate)

Content-based, or
context-based approach, or hybrid one?



- Application Layer

- Content complexity
- Quality of Experience
- User preference
- Required Latency
- Required Resources
- Required Scalability
- Required Availability
- Security and Privacy
- Fairness
- etc.

- Resource Layer

- Available resources (i.e., computation, storage, bandwidth)
- Performance
- Network Quality of Service (QoS)
- Cost
- etc.



- Video Contribution

- Per-title encoding
- Encoding parameters
- Bitrate Ladder
- Segment size
- etc.

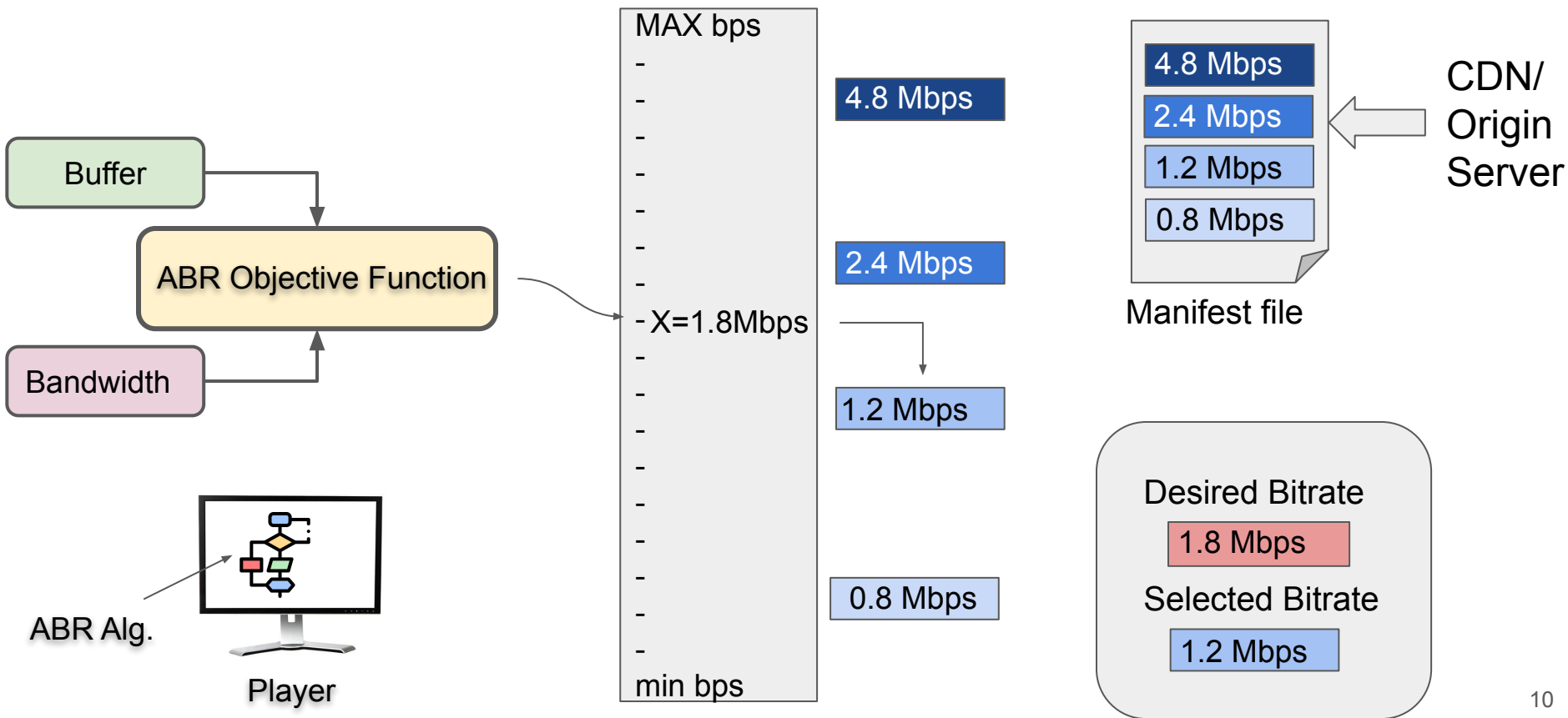
- Video Distribution

- Data path between streamer and encoder
- Multi-path data transmission between MEC server
- Function placement (i.e., VNF (transcoding))
- Cache Management
- Utilization

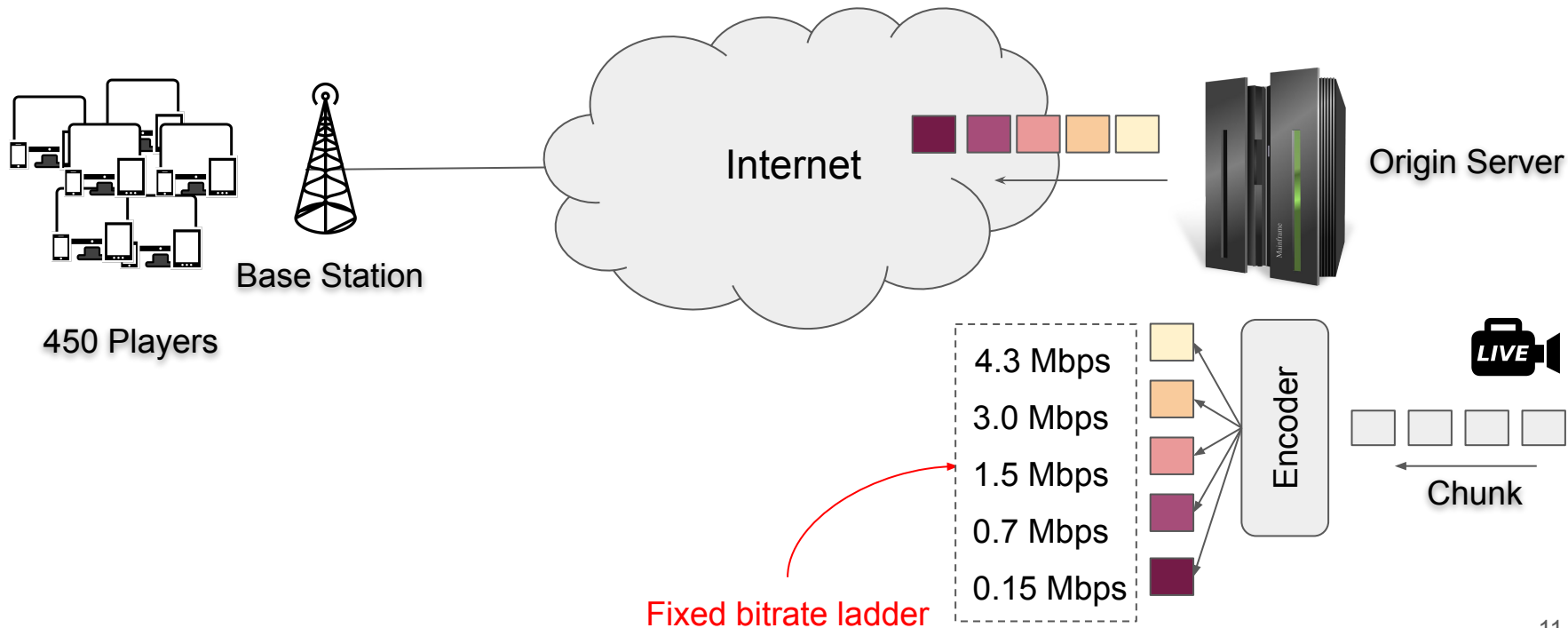
- Video Consumption

- ABR Algorithm design
- Super resolution
- etc.

Selected Bitrate vs. Desired Bitrate



Motivating Example



Motivating Example

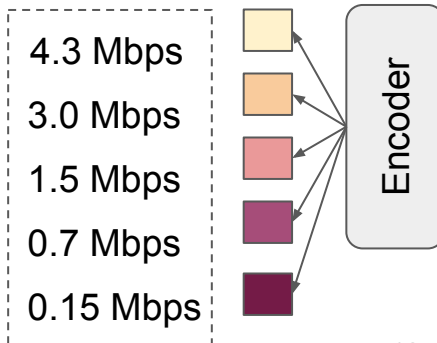


450 Players

Selected bitrate
50 players → 4.3
80 players → 3.0
150 players → 1.5
120 players → 0.7
50 players → 0.15



Origin Server



Motivating Example

VMAF: Video Multimethod Assessment Fusion
(full-reference video quality metric)

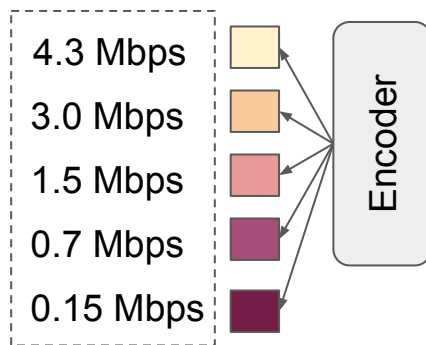


450 Players

Selected bitrate	Desired bitrate	Quality (VMAF)
50 players → 4.3	50 players → 4.3	98
80 players → 3.0	50 players → 3.2	97
	30 players → 3.0	96.5
150 players → 1.5	50 players → 2.5	96
	100 players → 2.0	95
	0 players → 1.5	80
120 players → 0.7	100 players → 1.0	75
	20 players → 0.7	65
50 players → 0.15	50 players → 0.15	20



Origin Server



Motivating Example

Selected bitrate	Desired bitrate	Quality
50 players → 4.3	50 players → 4.3	98
80 players → 3.0	50 players → 3.2	97
	30 players → 3.0	96.5
150 players → 1.5	50 players → 2.5	96
	100 players → 2.0	95
	0 players → 1.5	80
120 players → 0.7	100 players → 1.0	75
	20 players → 0.7	65
50 players → 0.15	50 players → 0.15	20

LALISA
L=5

4.3 Mbps
2.0 Mbps
1.0 Mbps
0.7 Mbps
0.15 Mbps

LALISA
L=4

2.0 Mbps
1.0 Mbps
0.7 Mbps
0.15 Mbps

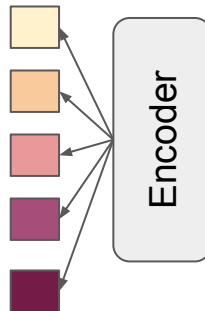
LALISA
L=3

2.0 Mbps
1.0 Mbps
0.15 Mbps

LALISA
L=2

1.0 Mbps
0.15 Mbps

4.3 Mbps
3.0 Mbps
1.5 Mbps
0.7 Mbps
0.15 Mbps

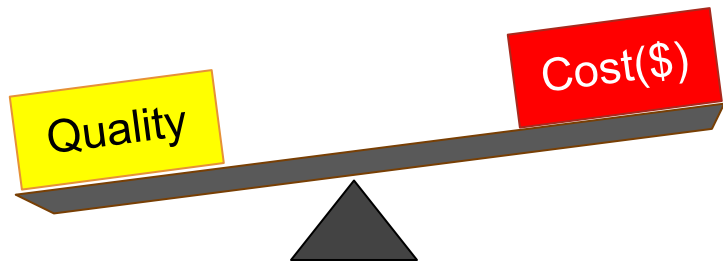


Origin Server

Motivating Example

VMAF and bandwidth comparison

	Avg. VMAF	Avg. BW (Mbps)
Fixed	74.2	1.7
LALISA (L=5)	81.5	2.0
LALISA (L=4)	80.8	1.5
LALISA (L=3)	78.8	1.4
LALISA (L=2)	66.4	0.8



LALISA
L=5

4.3 Mbps
2.0 Mbps
1.0 Mbps
0.7 Mbps
0.15 Mbps

LALISA
L=4

2.0 Mbps
1.0 Mbps
0.7 Mbps
0.15 Mbps

LALISA
L=3

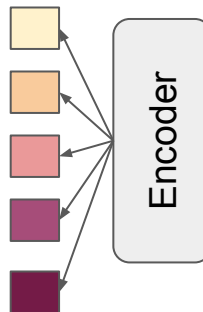
2.0 Mbps
1.0 Mbps
0.15 Mbps

LALISA
L=2

1.0 Mbps
0.15 Mbps

Fixed

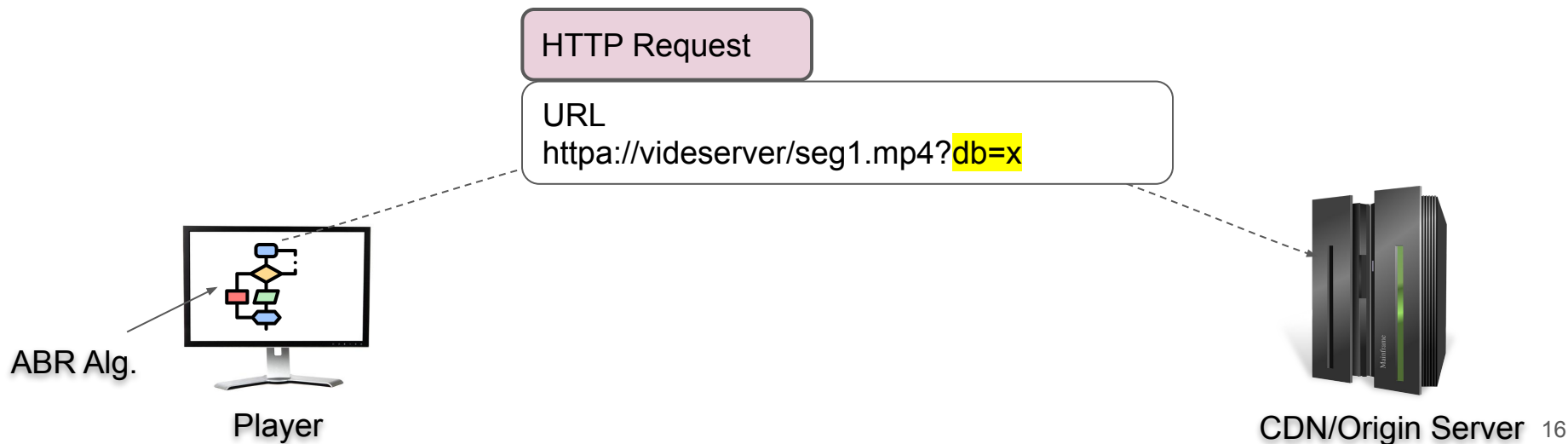
4.3 Mbps
3.0 Mbps
1.5 Mbps
0.7 Mbps
0.15 Mbps



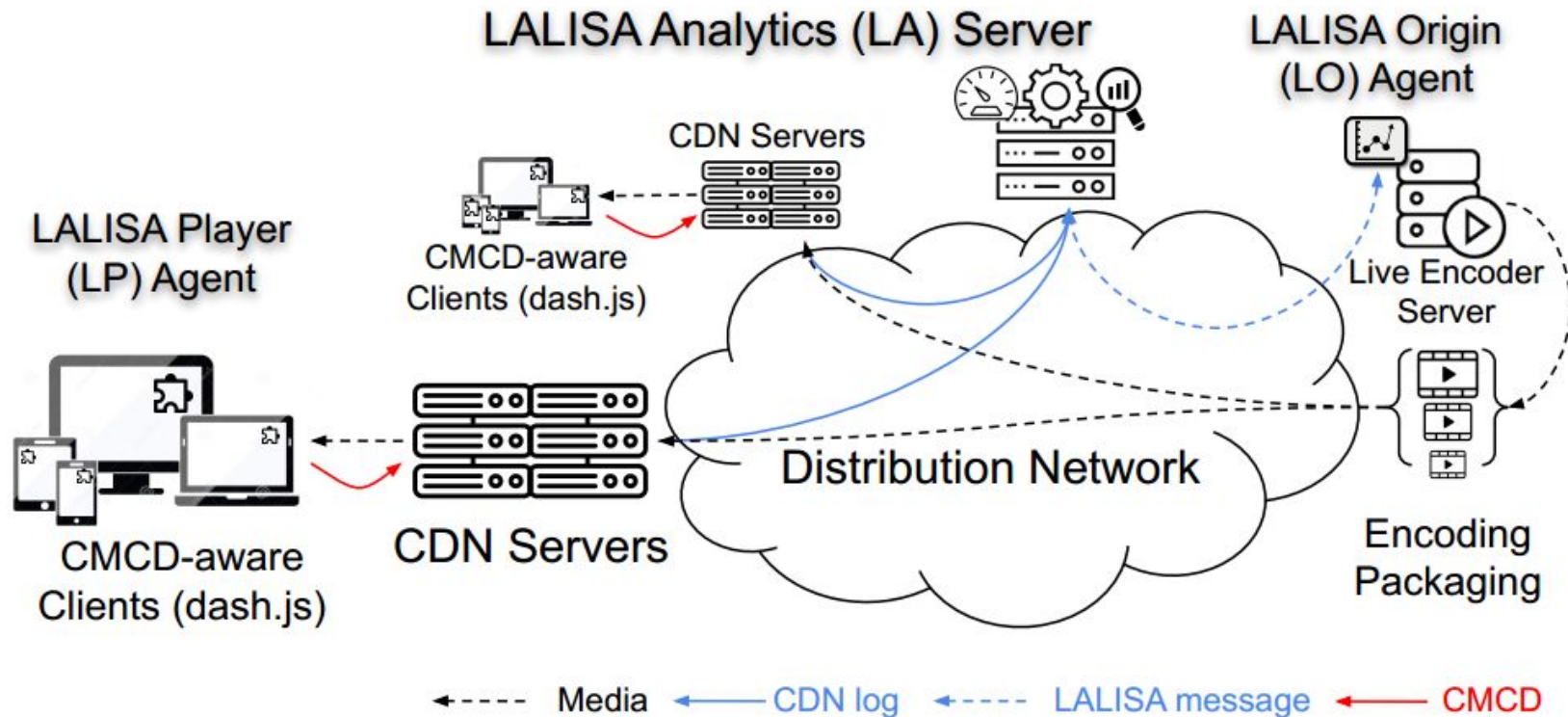
Origin Server

CMCD: Common-Media-Client-Data

How does a player send its desired bitrate to the server?



LALISA Architecture



LALISA Player (LP) Agent

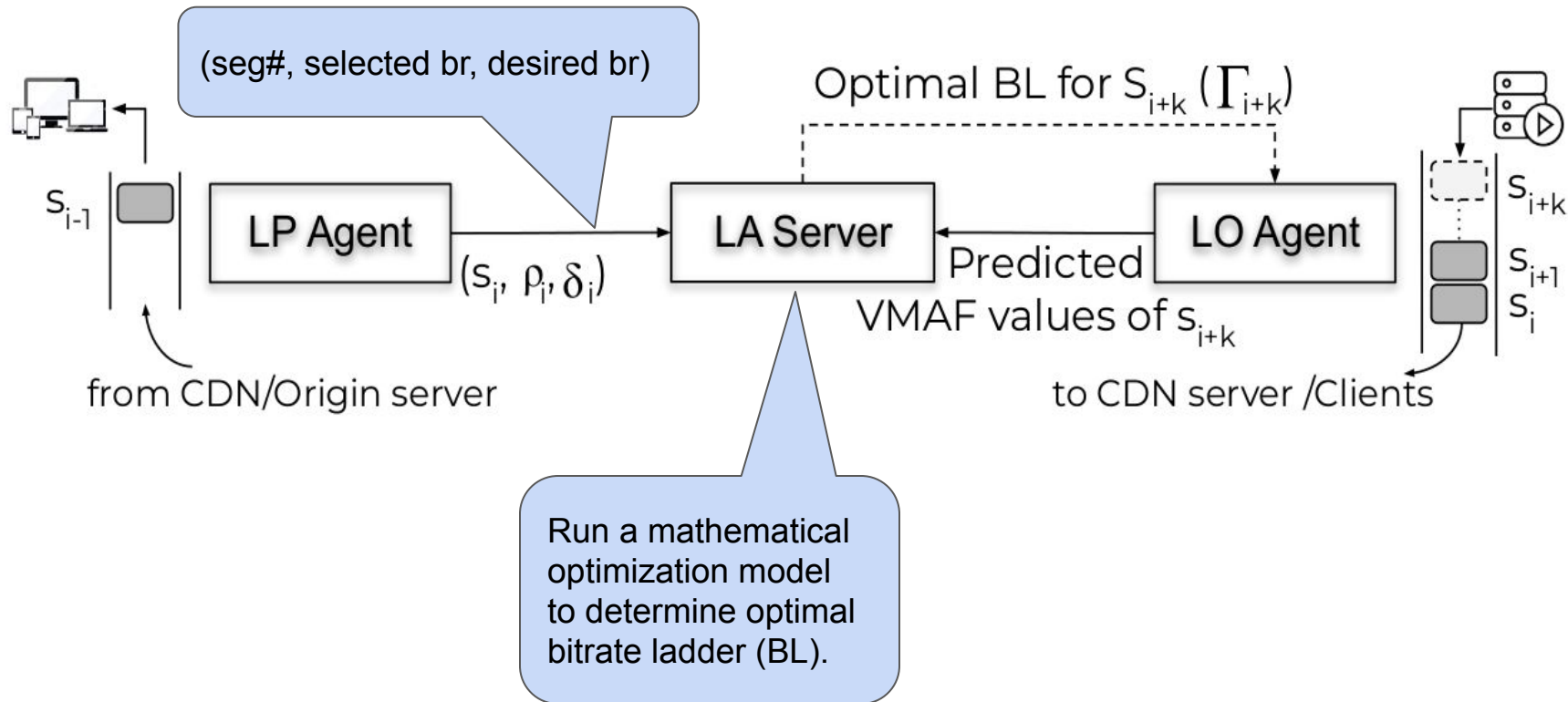
```
1  const lalisaGetDesiredBitrate = (lalisaParams,
   lalisaUtility, throughput, bufferLevel, ABR,
   possibleBitrate) => {
2  let desiredBitrate, score;
3  if (ABR == 'abrThroughput') {
4  for (let i = possibleBitrate.length - 1; i >= 0; i
   --) {
5      if (throughput >= possibleBitrate[i]) {
6          desiredBitrate= possibleBitrate[i];
7          break;}
8      }
9  }
10 else if (ABR == 'abrBola') {
11 for (let i = 0; i < possibleBitrate.length; ++i) {
12     let calculatedScore = (lalisaParams.Vp *
13     (lalisaUtility[i] + lalisaParams.gp)
14     - bufferLevel) / possibleBitrate[i];
15     if (!score || calculatedScore >= score) {
16         score = calculatedScore;
17         desiredBitrate = possibleBitrate[i];
18     }
19 }
20 return desiredBitrate;}
```

The LP agent is actually a **plug-in** that can be deployed with a HAS client to assist the ABR algorithm in **extracting the desired bitrates**.



dash.js

LALISA Analytics (LA) Server



Maximize $a \times \text{Quality} - (1-a) \times \text{Bitrate}$

Each request should be served with one Bitrate

$$\text{Maximize} \quad \alpha \times \sum_{g \in G} (Q_g - q_g) / \Pi - \beta \times B / \pi$$

$$x_g + \sum_{j \in G \& j < g} y_{g,j} = 1, \quad \forall g \in G$$

$$\sum_{g \in G \& j < g} y_{g,j} \leq x_j \times \lambda, \quad \forall j \in G$$

Maximum generated data regarding selected bitrates

$$\sum_{g \in G} x_g \times \delta_g \leq B$$

Amount of VMAF improvement by serving desired bitrates

$$(U(\delta_g) - U(\rho_g)) \times x_g \times n_g = Q_g \quad \forall g \in G$$

Amount of VMAF degradation by serving with lower bitrates

$$(U(\rho_g) - U(j)) \times y_{g,j} \times n_g = q_g \quad \forall g \in G, j \leq \rho_g$$

Maximum length of the bitrate ladder

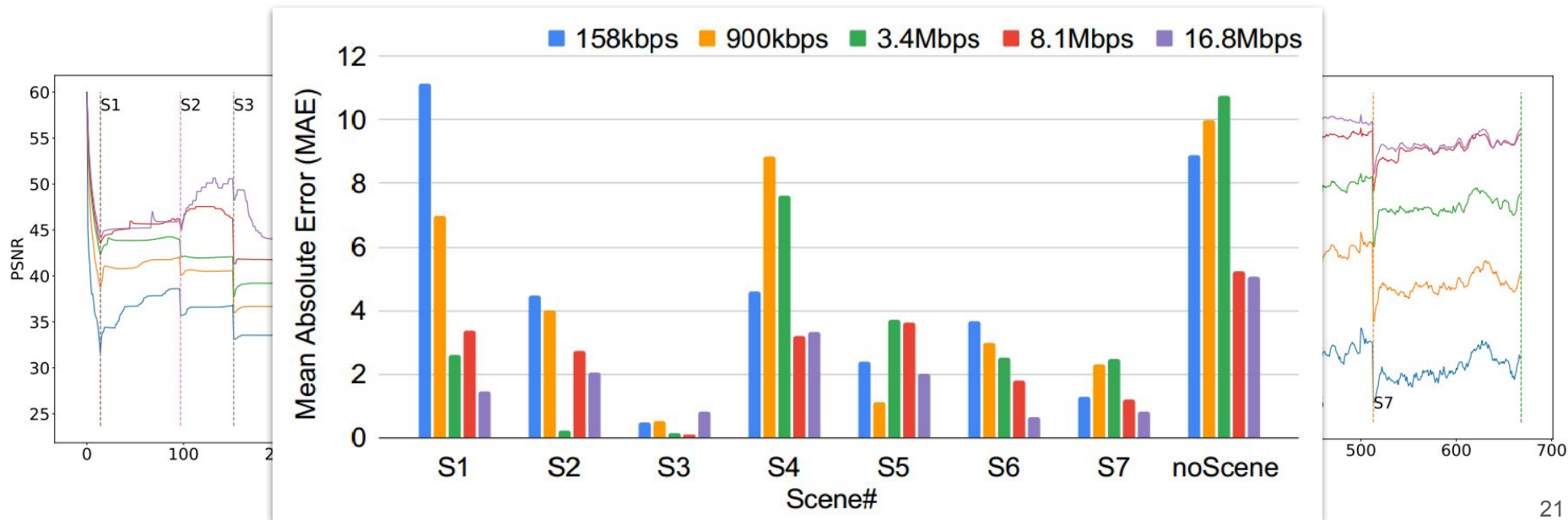
$$\sum_{g \in G} x_g \leq \ell$$

Variables:

$$Q_g, q_g, B \geq 0, \text{ and } x_g, y_{g,j} \in \{0, 1\}$$

LALISA uses the scene-cut detection algorithm

Result of running scene-cut detection algorithm and measured mean absolute error of the proposed VMAF prediction algorithm



LALISA testbed

- Segment duration:
 - 1, 2, and 4 seconds
- Experiment time:
 - 120 seconds
- Network trace:
 - LTE (player), Fixed (origin)
- Codec:
 - H.264
- Content:
 - Animation
- Player#:
 - 10, 20, 40

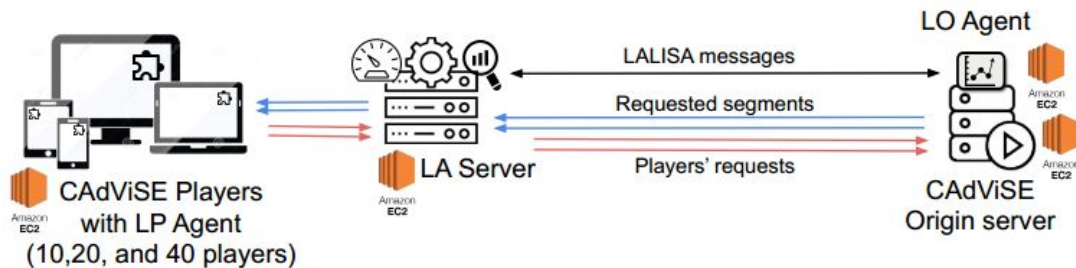
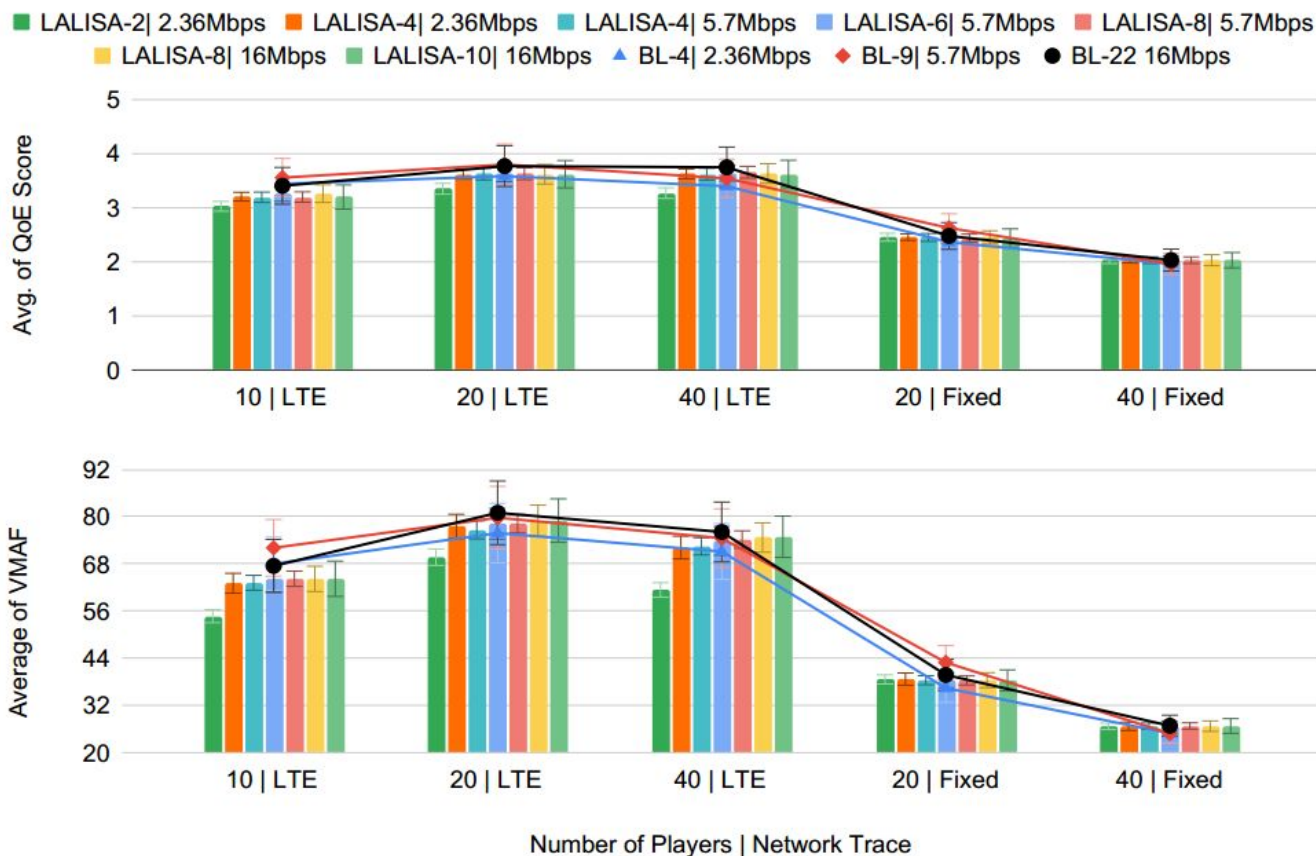
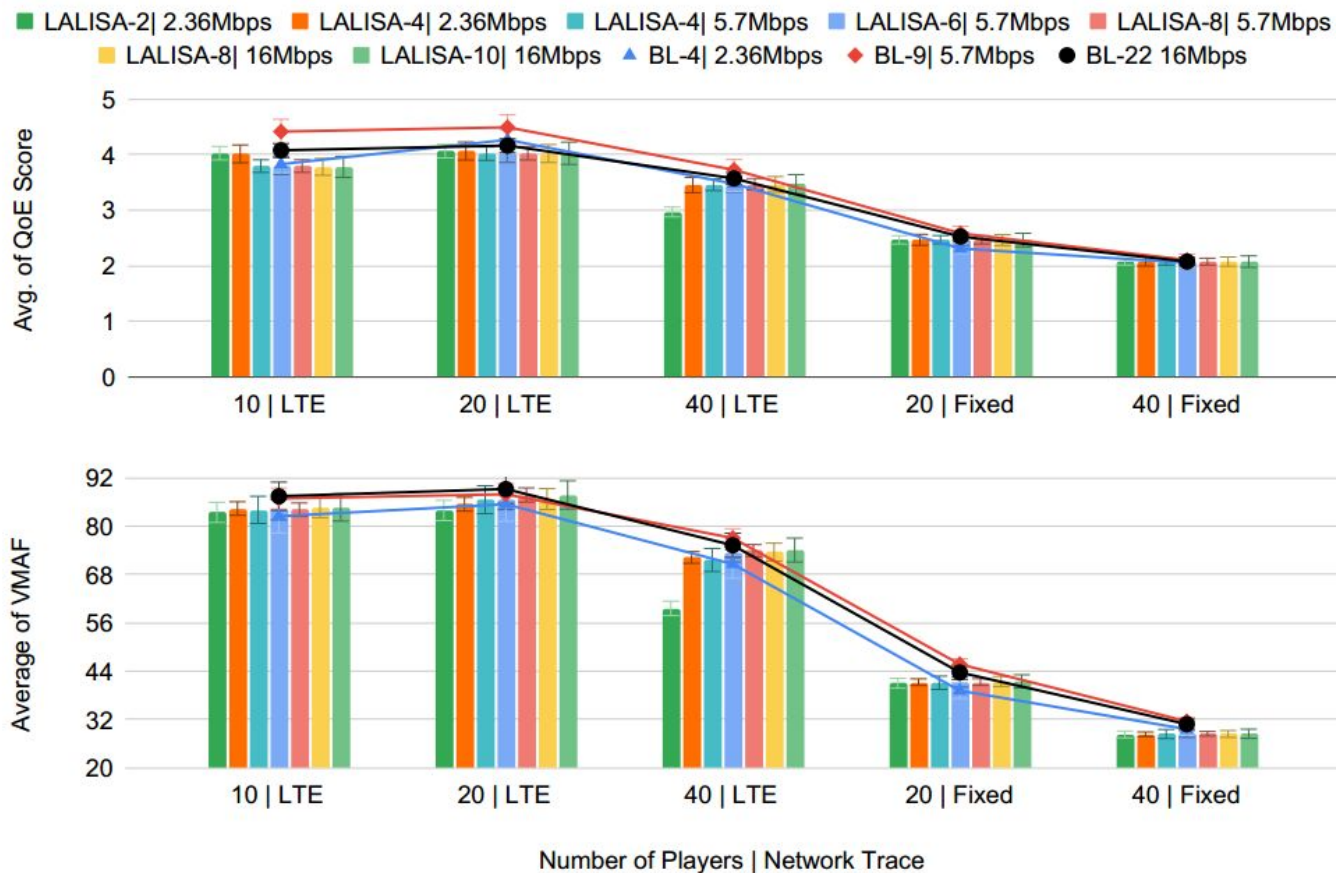


Table I: Bitrate ladders (BLs) used in the experiments.

BL	Res.	Bitrates (kbps)	Res.	Bitrates (kbps)
BL-4	180p	158	288p	750
	540p	1760	720p	2360
BL-9	180p	158, 293	288p	600
	360p	893	540p	1760
	720p	2360, 3500	1080p	4300, 5700
BL-22	180p	158, 240, 293	216p	375, 453
	288p	570, 600, 750	360p	893, 1000, 1400
	540p	1760	720p	2000, 2300, 3000, 3500
	1080p	4300, 5000, 5700	1440p	8000, 11000, 16000



Measured QoE score and VMAF values for different numbers of clients and network traces with 1s segment duration



Measured QoE score and VMAF values for different numbers of clients and network traces with 2s segment duration

Impact of LALISA on Bandwidth and Computation Cost

Table II: Average % (over all segment durations) of improvement and degradation of LALISA vs. BL-9.

LALISA-	BL Size Decrease(%)	Degradation in QoE (%) — Δ VMAF							
		LTE				Fixed			
		Bola	Throughput	LoL	BBA0	Bola	Throughput	LoL	BBA0
4 5.7Mbps	55.6	6.2 2.6	0.8 1.1	10 3.9	2.8 2.8	4.1 6.4	1.6 2.4	0.2 1.2	1.2 1.0
6 5.7Mbps	33.3	5.3 1.5	0.3 0.5	10 3.3	1.3 1.6	4.1 6.3	1.3 2.3	0.2 0.8	0.0 0.8
8 5.7Mbps	11.1	5.3 1.0	0.1 0.0	7.0 3.0	1.0 0.4	4.3 6.3	1.0 2.0	0.1 0.1	0.0 0.0
Average	33.3	5.6 1.7	0.4 0.5	9.0 3.4	1.7 1.6	4.1 6.3	1.3 2.2	0.1 0.7	0.4 0.6

Table I: Bitrate ladders (BLs) used in the experiments.

BL	Res.	Bitrates (kbps)	Res.	Bitrates (kbps)
BL-9	180p	158, 293	288p	600
	360p	893	540p	1760
	720p	2360, 3500	1080p	4300, 5700

LALISA-	BL Size Decrease(%)	BW. Cost Saving (%)	Comp. Cost Saving (%)
4 5.7Mbps	55.6	29.5	51.3
6 5.7Mbps	33.3	15.0	13.2
8 5.7Mbps	11.1	10.3	08.6
Average	33.3	18.2	24.4

Conclusion

- LALISA : an end-to-end bitrate ladder optimization approach
 - LALISA architecture
 - Desired bitrate vs. selected bitrate
 - Mixed integer linear programming optimization model
 - Evaluated LALISA in various scenarios
 - LALISA outperformed fixed-length ladders in terms of QoE and resource consumption
- Future work:
 - Consider different players and content types.
 - Improve LALISA with applying ML techniques instead of MILP



LALISA: Adaptive Bitrate Ladder Optimization in HTTP-based Adaptive Live Streaming

thank you



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